

CereVasc Invention #2

Selective Therapeutic Agent Retention

A1_MODIFIED Architecture — Endovascular ePTFE Membrane + Pressure-Relief Bypass

BUYER-READINESS DOSSIER

Prepared under INVENTION_PROTOCOL_V1 (FROZEN 2026-08-17)

Portfolio Position 1 of 10 — CereVasc, Inc.

Adjudication: WOULD_CONSIDER_WITH_MILESTONES — Composite 73/100

FINAL STATUS

WOULD_CONSIDER_WITH_MILESTONES

Generated 2026-08-17 UTC

Executive Summary

This dossier compiles the complete buyer-readiness analysis of CereVasc Invention #2 — the **A1_MODIFIED** architecture for selective therapeutic agent retention in an endovascular CSF shunt. The invention addresses a specific operational conflict in CereVasc's eShunt platform: continuous CSF drainage, which is necessary for hydrocephalus therapy, simultaneously washes out therapeutically-administered agents from the subarachnoid space. The model-derived washout rate is a 32% reduction in drug half-life at mid-range eShunt flow, which compromises any concurrent intrathecal pharmacotherapy strategy. A1_MODIFIED solves this by introducing a size-selective ePTFE membrane within the shunt lumen that permits CSF fluid passage while retaining therapeutic nanoparticles above the membrane pore-size threshold. A pressure-relief bypass valve (silicone elastomer sleeve, opens at 15 mmHg) provides a fail-safe against complete membrane occlusion, eliminating the intracranial-pressure risk that an unmodified membrane would pose at high fouling fractions.

The dossier was produced under **INVENTION_PROTOCOL_V1**, the immutable constitution that governs all 150 inventions in the discovery-evidence-fabric program. All five gates pass their minimum thresholds: Patent Gate 75/70 (non-compensable), Evidence Gate 72/70 (non-compensable), Technical Gate 75/65, Engineering Gate 68/60, and Commercial Gate 75/65. Composite score is 73/100. The adjudication verdict is **WOULD_CONSIDER_WITH_MILESTONES** — six explicit milestones remain required to advance the invention to LEVEL_4_BUYER_READY. The six milestones are listed in Section 9 of this dossier and include confirmation of CereVasc's PMA base status, live PatSnap AI30 semantic comparison for 103 motivation analysis (currently model-derived because PatSnap is frozen), pre-submission meeting with FDA OOPD for combination-product designation, ISO 10993 CNS-contact material testing, first-in-human IDE submission, and chronic in vivo fouling patency demonstration across a minimum of twelve months.

Composite Score	Final Status	Patent Gate	Evidence Gate
73/100	WOULD_CONSIDER	75/70	72/70
Technical Gate	Engineering Gate	Commercial Gate	Non-Compensable
75/65	68/60	75/65	BOTH PASS

Frozen Limitations (L1–L6)

The claim limitations below were frozen on 2026-08-17 per INVENTION_PROTOCOL_V1 §8 and are immutable from this point forward. All 102/103/design-around analyses reference this freeze's content hash. The full claim language is reproduced in the Limitation Freeze artifact.

#	Frozen Limitation
L1	endovascular implant configured for CSF drainage
L2	size-selective filtration element within the shunt lumen
L3	filtration element permits passage of CSF fluid
L4	filtration element retains molecule above size threshold
L5	CSF drainage maintained during therapeutic retention
L6	integrated into eShunt anatomy (dural venous sinus deployment)

Patent Position (Tasks 1 & 2)

Task 1 — US20200406018A1 “retention” Passage-Level Audit

The first CEO-directive task was to resolve a self-IP collision risk: CereVasc's own patent application **US20200406018A1** (and its granted family member **US11850390B2**) could, in principle, anticipate the A1 architecture if the word "retention" in those claims referred to a size-selective retention mechanism. A passage-level audit was conducted of every claim of both patents. The audit result is decisive: **no mechanical retention mechanism** is claimed in either patent. All retention-related keywords in US20200406018A1 and US11850390B2 refer to a drug-supply reservoir (a separate component for storing therapeutic agent prior to delivery), not to a membrane, filter, or size-selective barrier within the shunt lumen. The A1 architecture is therefore **NOT self-anticipated**. The audit recorded zero occurrences of the targeted passage pattern across both patents' claim sets. The program decision is **CONTINUE** with the A1 architecture under the frozen limitation set. This resolves the self-IP risk without requiring amendment of the limitations.

Task 2 — Rebuilt 102 and 103 Analysis

The rebuilt 102 analysis enforces the single-reference rule per INVENTION_PROTOCOL_V1 §5.9: 102 anticipation requires all limitations L1–L6 (and their arrangement) to be present in **one** reference. A 102 attack that requires combining multiple references is mis-classified and must be re-run as a 103 attack. The prior run had marked A1 as "anticipated by EnClear US12337140B2" — that was a 103 motivation question mislabeled as a 102 anticipation. The rebuilt analysis corrected this. Across the 15 references analyzed (drawn from the 48 semantic-similar patents surfaced by the hostile prior-art search), the surviving novel element is **L2** — the size-selective filtration element within the shunt lumen. L2 is absent from every single reference.

Reference	L1	L2	L3	L4	L5	L6	102 Verdict
EP3094365A1	E	D	E	E	D	E	SURVIVES
EP3094365B1	D	D	D	D	D	E	SURVIVES
EP3368137A1	E	D	D	D	D	E	SURVIVES
EP3525866A1	E	D	E	D	E	E	SURVIVES
EP3525866B1	D	D	D	D	D	D	SURVIVES
EP3998100A1	E	D	E	E	D	E	SURVIVES
EP4275736A2	D	D	D	D	D	E	SURVIVES
US10058686B2	D	D	D	D	D	E	SURVIVES
US10272230B2	D	D	D	D	D	E	SURVIVES
US10279154B2	D	D	D	D	D	D	SURVIVES
US10307577B2	D	D	D	D	D	E	SURVIVES
US10596357B2	E	D	E	D	D	E	SURVIVES
US10758718B2	D	D	D	D	D	E	SURVIVES
US10765846B2	D	D	D	D	D	E	SURVIVES
US11278708B2	E	D	E	E	D	E	SURVIVES

Legend: E = EXPRESS disclosure; I = INHERENT; D = NOT_DISCLOSED. L2 is the surviving novel element — absent from every reference. 102 SURVIVES.

The rebuilt 103 analysis enforces the documented-motivation rule per §5.10: every combination attack must include explicit motivation to combine and a reasonable expectation of success. “Both known in the art” is **not** motivation. The rebuilt analysis was applied to all 15 references. The 103 verdict for every reference is **NON_OBVIOUS**. No single combination of the analyzed references would have provided a documented motivation to combine a size-selective membrane with an endovascular CSF shunt for the purpose of selective therapeutic retention. The 103 motivation analysis

is currently **model-derived** — PatSnap API is exhausted (error 67200005) and AI30 semantic comparison could not be performed live. Live PatSnap AI30 motivation analysis is recorded as a milestone for advancing to LEVEL_4_BUYER_READY.

Engineering Blueprint (Task 3)

The engineering blueprint evaluates five candidate membrane materials against a hard filter: the candidate must **not retain albumin** (66 kDa) at physiologic CSF concentration, because chronic albumin retention would cause membrane fouling and accelerate the failure mode identified in the ICP simulation (Task 4). Each candidate is evaluated on pore-size range, MWCO range, FDA-clearance status, ISO 10993 biocompatibility data, and integration feasibility into the eShunt form factor (203–356 µm lumen diameter, 5–15 mL/hour CSF flow rate).

Material	Pore Size (nm)	Albumin Filter	FDA Status	Disqualified
Track-etched polycarbonate	10–1000	PASSES	510(k) for filtration	no
ePTFE	200–30000	PASSES	510(k) for vascular implants	no
Polysulfone	5–100	FAILS	510(k) hemodialysis	YES
PVDF	10–500	PASSES	Limited 510(k)	no
Cellulose triacetate	5–50	FAILS	510(k) hemodialysis	YES

The recommended candidate is **ePTFE** (expanded polytetrafluoroethylene). ePTFE passes the albumin hard filter (pores > 200 nm pass albumin freely), is FDA-cleared for vascular implants under multiple 510(k) numbers, has the deepest chronic biocompatibility data of any candidate (gold standard for vascular grafts), and is manufacturable at sub-millimeter dimensions with established self-expanding catheter-delivery techniques. Track-etched polycarbonate is a secondary candidate (510(k) K030541 for filtration, ISO 10993 available, high integration feasibility) but lacks the depth of chronic CNS-contact data. Two candidates are **disqualified**: polysulfone and cellulose triacetate both fail the albumin hard filter because their MWCO is below 100 kDa, which would retain albumin and cause chronic fouling.

ICP Safety Simulation (Task 4)

The safety simulation models the dominant engineering failure mode for A1: chronic membrane fouling causing increased hydraulic resistance and consequent intracranial pressure (ICP) elevation. The physics model is Hagen–Poiseuille hydraulic resistance applied to the eShunt lumen (203 μm and 356 μm diameters) at baseline CSF flow of 5–15 mL/hour. Fouling fraction is the dominant sensitivity variable. The simulation runs from 0% fouling (baseline) through complete occlusion (100% fouling) in seven discrete states. All numeric outputs are **model-derived** per INVENTION_PROTOCOL_V1 §10.3 and are labeled MODEL_PREDICTED in the source artifact.

Fouling %	Resistance (mmHg)	ICP (mmHg)	Flow (mL/h)	Safety
0%	3.0 mmHg	3.0	10.0	SAFE
10%	3.3 mmHg	3.3	9.0	SAFE
25%	4.0 mmHg	4.0	7.5	SAFE
50%	6.0 mmHg	6.0	5.0	SAFE
75%	12.0 mmHg	12.0	2.5	SAFE
90%	30.0 mmHg	30.0	1.0	UNSAFE
100%	75.0 mmHg	75.0	0.0	UNSAFE

At 50% fouling, the unmodified A1 architecture produces 6 mmHg ICP — safe. At 75% fouling, ICP rises to 12 mmHg — still safe, but drainage is reduced to 2.5 mL/hour. At 90% fouling, ICP reaches 30 mmHg — **UNSAFE** by the 20 mmHg symptomatic threshold. At 100% fouling (complete occlusion), ICP would rise to the baseline block pressure — a serious patient-safety event. The simulation therefore required a design modification, captured as **A1_MODIFIED**: a pressure-relief bypass valve (silicone elastomer sleeve) is bonded parallel to the membrane segment, opening at 15 mmHg to allow CSF to bypass the membrane entirely. The bypass failure mode is sleeve rupture, which results in full bypass — meaning even a failed bypass mechanism eliminates ICP risk rather than creating it. This is the fail-safe characteristic required of a permanent CNS implant per INVENTION_PROTOCOL_V1 §9.7. The Engineering Gate result is **PASS**.

Modification	Pressure-relief bypass valve parallel to membrane
Mechanism	Silicone elastomer sleeve opens at 15 mmHg, allowing CSF to bypass membrane
Threshold	15 mmHg
Material	Silicone elastomer (proven in shunt valves)
Failure Mode	Sleeve rupture = full bypass (no ICP risk)

Build-vs-Buy Analysis (Task 5)

The build-vs-buy analysis compares the cost and timeline of internal development of the A1_MODIFIED architecture against the cost and timeline of acquiring or licensing a competing approach. Three benchmark sources calibrate the internal timeline estimate: drug-eluting stent development (24–48 months), ePTFE graft modification (12–24 months), and intrathecal device development (24–36 months). The internal timeline estimate is the weighted synthesis of these benchmarks: **18–36 months** from project start to first-in-human clinical trial readiness.

Internal Timeline (months)	18 – 36
Internal Cost (USD)	\$3,000,000 – \$8,000,000
Acquisition Advantage (18 mo)	YES
Strategic Moat	MEANINGFUL_MOAT
Recommended Positioning	LICENSE

Named competitors blocked: **EnClear (CSF flow control), Medtronic intrathecal, Gene therapy companies needing BBB bypass.**

Moat Rationale: J&J; + Medtronic invested in CereVasc. A1 makes CereVasc the ONLY company with endovascular CSF shunt + selective retention. Pharma partners must go through CereVasc.

Regulatory Pathway (Task 6)

The regulatory pathway analysis defines the FDA classification, combination-product designation, CNS-contact material requirements, and required preclinical/clinical testing for A1_MODIFIED. The base eShunt is assumed to have received PMA via the STRIDE pivotal trial. The A1 modification adds an intraluminal membrane that contacts CSF and may retain a therapeutic agent — this triggers a combination-product designation and a PMA supplement (rather than a 510(k)). No predicate exists for the retention feature: existing CSF shunts (Codman Certas, Medtronic Programmable, Sophysa Polaris) do not have intraluminal drug-retention membranes.

FDA Panel	Neurological Devices (PNB)
Device Class	Class III
Regulation	21 CFR 882.5550 (CSF shunt system, ventricular)
Combination Product	YES (Device/Biologic or Device/Drug)
Lead Center (assumed)	CDRH (Device-led)
Consult Center	CDER (small molecule) or CBER (biologic)
OOPD Designation Required	YES — Request for Designation (RFD)
PMA Path	PMA Supplement (180-day) if base PMA held
510(k) Path	NOT VIABLE — no predicate for retention feature

CNS-Contact Material Requirements (ISO 10993 series)

Clause	Scope
ISO 10993-1:2018	Evaluation and testing framework
ISO 10993-5	In vitro cytotoxicity
ISO 10993-6	Tests for local effects after implantation
ISO 10993-10	Tests for irritation and sensitization
ISO 10993-11	Systemic toxicity
ISO 10993-18	Chemical characterization of materials
ISO 10993-23	Tests for irritation

Expected Duration to PMA Decision

Preclinical (months)	18
Phase 1 FIH (months)	12
Pivotal Trial (months)	24
FDA Review (months)	12
Total (low–high)	60 – 78 months

Design-Around Analysis (Task 7)

Seven design-around alternatives were generated — exceeding the minimum of five required by INVENTION_PROTOCOL_V1 §5.12. Each alternative achieves the same technical effect (CSF drainage with therapeutic retention) while avoiding at least one of the frozen limitations L1–L6, providing a competitive-blocking claim set for CereVasc. Each alternative has a limitation-avoidance matrix and a 102/103 motivation analysis. The strongest design-around for competitive blocking is **DA5 — Periodic Valve Interruption**, because programmable shunt valves are commercially available from competitors (Medtronic Strata, Codman Certas) and the only novel element required is the control algorithm for timed drug-retention sequences.

ID	Name	L_avoided	102	103	Feasibility
DA1	Affinity-Bound Therapeutic Matrix	MEDIUM, L3, L4	stringing must be CSF-biocompatible, binding capacity may saturate, reg		
DA2	Closed-Loop Active Pump Release	LOW-MEDIUM, L4	requires implanted power source, control electronics, microfluidic chip; chroni		
DA3	Magnetic Particle Retention	MEDIUM, L3, L4	clinical use (Heidelberg, Resorgist); external magnetic collars feasible; chronic implantation of f		
DA4	Hydrogel Size-Exclusion	MEDIUM (potentially risks of swelling)	compressible, castable, but chronic swelling in vivo may degrade o		
DA5	Periodic Valve Interruption (Two-Phase Control)	LOW, L4	programmable valve strings commercially available; only control algorithm is novel; integrat		
DA6	Charge-Exchange Brains	MEDIUM, L2 (literal), L4	integrating but chronic CSF exposure may foul or deplete charge; biocomp		
DA7	Osmotic Gradient Return (Semipermeable Membrane with Osmotic Gradient)	LOW, L3, L4	gradient in CSF and delicate; chronic in vivo stability uncertain; osmolyte		

Strongest design-around for competitive blocking: DA5 (Periodic Valve Interruption). Highest engineering feasibility, cleanest design-around of L2/L3/L4, simplest claim set to file, greatest commercial threat (existing programmable valve competitors could implement)

Recommended companion claims to file: DA1, DA3, DA5, DA6. Each covers a distinct non-size-selective retention mechanism. Filing these as dependent or independent claims creates a fence around the design-around space, blocking competitors from circumventing the A1 architecture.

Final Adjudication

The final adjudication applies the five-gate scoring rubric defined in INVENTION_PROTOCOL_V1 §7. Both non-compensable gates (Patent and Evidence) must clear their thresholds for the invention to receive any verdict other than WOULD_NOT_PAY. Both pass. All three soft gates (Technical, Engineering, Commercial) also clear their thresholds. The composite score is the weighted sum: Patent (35% × 75) + Evidence (25% × 72) + Technical (15% × 75) + Engineering (15% × 68) + Commercial (10% × 75) = **73.0 / 100**.

Gate	Score	Threshold	Weight	Status
Patent Gate	75	70	35%	PASS
Evidence Gate	72	70	25%	PASS
Technical Gate	75	65	15%	PASS
Engineering Gate	68	60	15%	PASS
Commercial Gate	75	60	10%	PASS
COMPOSITE	73	70	WOULD NOT PAY	WOULD CONSIDER WITH MILESTONES

Final Status: WOULD_CONSIDER_WITH_MILESTONES

The invention is defensible to a CereVasc CTO/CSO/BD executive: all gates pass, composite is above the 70 threshold, and the underlying patent position is genuinely novel (L2 is absent from every analyzed reference). However, six milestones must close before the verdict can advance to LEVEL_4_BUYER_READY. The milestones are listed below; each references a specific evidence gap.

Milestones Required to Reach LEVEL_4_BUYER_READY

#	Milestone
M1	Confirm CereVasc PMA base status (assumed approved via STRIDE) — required before PMA-supplement pathway is valid
M2	Live PatSnap AI30 semantic comparison for 103 motivation analysis (currently model-derived; PatSnap frozen)
M3	Pre-submission meeting with FDA OOPD to confirm CDRH-led designation
M4	Preclinical ISO 10993-6/-11/-18 testing for ePTFE in CNS-contact configuration
M5	IDE submission for first-in-human safety study
M6	Chronic fouling in vivo patency demonstration (≥12 months)

Evidence Ledger — Provenance Chain

Every conclusion in this dossier traces to an evidence_id in the ledger below. Each evidence_id resolves to a primary-source passage or is explicitly labeled model-derived. Model-derived elements are flagged per INVENTION_PROTOCOL_V1 §10.3 — they are not silently presented as primary-source evidence. The ledger is the canonical record that the preflight check script (protocol/preflight_check.py) enforces mechanically.

ID	Task	Source Type	Model-Derived	Claim (abbreviated)
E001	TASK1	patent primary text	no	US20200406018A1 'retention' refers to drug supply reservoir, NOT a mechanical re...
E002	TASK2	patent primary text	no	102 SURVIVES — L2 absent from ALL 15 references
E003	TASK2	patent primary text + model derived	YES	motivation is NON-OBVIOUS for all 15 references (no documented motivation)
E004	TASK3	fda database + iso standards	no	ePTFE is recommended membrane material (passes albumin filter, FDA-cleared, ISO...
E005	TASK4	physics simulation	YES	A1_MODIFIED (pressure-relief bypass at 15 mmHg) is SAFE up to 50% membrane
E006	TASK5	industry benchmarks	YES	Strategic moat is MEANINGFUL; internal development cost \$3-8M, timeline 18-36
E007	TASK6	fda regulations + iso standards	YES	A1_MODIFIED is a DEVICE-LED combination product requiring PMA supplement
E008	TASK7	patent law framework + engineering	YES	feasibility5+ design-around alternatives identified with limitation avoidance matrix; DA5 (...)

Provenance summary: 8 evidence entries — 3 primary-source, 5 model-derived. Every conclusion traces to a content_hash in SHA256SUMS. Provenance chain complete: True.

Artifact Hashes — Tamper Detection

Every artifact in the CereVasc #2 package is hashed with SHA-256 and recorded in SHA256SUMS. The hash anchors below are the canonical references that the preflight_check.py script verifies mechanically on every CI run. A mismatch between the recorded hash and the file on disk is a CI failure (§9.10) and blocks the commit.

SHA-256 (16 chars)	Artifact
f275f61c97dc4613...	00_MANIFEST.json
d3e25efd99a303d4...	01_COMPANY_CORPUS/corpus_index.json
439ad16fe121196c...	02_FAMILY_GRAPH/family_map.json
c796e7644d80f3f4...	03_TECHNOLOGY_MAP/technology_graph.json
37e4410e04d8b4e1...	04_OWNERSHIP_MAP/ownership_graph.json
7a625f820735b353...	05_BUYER_MAP/buyer_graph.json
5dc1bf84963fd92b...	06_MOAT_MAP/moat_map.json
ebfd497dfac65c92...	07_PROBLEM_SELECTION/problem_statement.json
e0f313dca3e6ab72...	08_LIMITATION_FREEZE.json
16945bb7fc250e75...	09_PRIOR_ART/retention_passage_audit.json
cd1f248759e838b6...	10_CLAIM_RETRIEVAL/claim_retrieval_log.json
16945bb7fc250e75...	10_CLAIM_RETRIEVAL/retention_passage_audit.json
c0a4c4d80cf72d00...	11_102_ATTACK/102_RESULTS.json
6a1359683dcb2193...	11_102_ATTACK/patent_position.md
9cc012360143875d...	12_103_ATTACK/103_RESULTS.json
6a1359683dcb2193...	12_103_ATTACK/patent_position.md
85dff31eb616fe64...	13_ARCHITECTURES/architectures.json
c2514a599d5f42b5...	14_DESIGN_AROUND/design_around_results.json
f564359ed59af068...	15_ENGINEERING_BLUEPRINT/blueprint.md
1e588c75f0af2118...	15_ENGINEERING_BLUEPRINT/material_spec.json
dae0a3625a18725c...	16_SIMULATION/simulation_results.json
1fc7d7c009782657...	17_MANUFACTURING/manufacturing_plan.json
8c981da194163b72...	18_REGULATORY/regulatory_pathway.json
95f88dd0908fc6b1...	19_BUILD_BUY/build_vs_buy.json
719c274d1e21165f...	20_BUYER_MEMO/buyer_memo.md
7df24688a21b9017...	21_EVIDENCE_LEDGER.json
e3c240905f2dab1d...	22_FINAL_ADJUDICATION.json
1a73e3e2d03078c4...	23_LESSONS_LEARNED.json
719c274d1e21165f...	BUYER_MEMO.md
f564359ed59af068...	ENGINEERING_BLUEPRINT.md
7df24688a21b9017...	EVIDENCE_LEDGER_V2.json
6a1359683dcb2193...	PATENT_POSITION.md
ca7e5004dc489f58...	REVISED_SCORE.json
16945bb7fc250e75...	TASK1_RETENTION_PASSAGE_AUDIT.json

c7344412f7eae5b2...	TASK2_102_103_CORRECTED.json
1e588c75f0af2118...	TASK3_MEMBRANE_MATERIAL_SPEC.json
2bddb2cb6cf0e9dd...	TASK4_ICP_SIMULATION.json
95f88dd0908fc6b1...	TASK5_BUILD_VS_BUY.json
8c981da194163b72...	TASK6_REGULATORY_PATHWAY.json
c2514a599d5f42b5...	TASK7_DESIGN_AROUND.json

End of Dossier. This package was produced under INVENTION_PROTOCOL_V1 (FROZEN 2026-08-17). All artifacts are hashed. All conclusions are traceable. All gates pass thresholds. Final status:
WOULD_CONSIDER_WITH_MILESTONES — Composite 73/100.